

UNIVERSITY COLLEGE LONDON, Department of Economics
MRes Dissertation Proposal

Complete this form and email it to Daniella Harper daniella.harper@ucl.ac.uk by Monday 1 April 2026.

1. Name of Student: Runbin Wang
2. Title of Proposed Dissertation: Structural Dynamics and Asymmetric Pass-Through of Fuel Tax Intervention
3. Name of dissertation Supervisor: Dr. Joao Granja
4. Are you intending to join the MPhil/PhD Programme? Yes

If you answered yes to question 4, note the following: While the department does not expect you to have finished their dissertation work by August, you should aim to have produced some written material as evidence of your progress to your supervisor by 7th August. In addition, your supervisor will be required to provide the following information.

- ☐ If the work, they have seen so far meets the expectations of admission to the MPhil/PhD.
- ☐ If they predict a dissertation mark of at least 60% or above based on current information.
- ☐ If they support admission to the MPhil Programme.
- ☐ If they can continue supervising.

Give the dates for four MRes dissertation supervises meetings below. These meetings are to take place during the summer, usually in June, twice in July and in August. The meetings can take place online.

1. Date of first meeting (June): 23rd Jun
2. Date of second meeting (early July): 7th Jul
3. Date of third meeting (mid - July): 21th Jul
4. Date of fourth Meetings (August 1st-7th): 4th Aug

Supervisor's signature:



Student's signature:



- ☒ I have attached my research proposal (1-2 pages will suffice).
- ☒ I have made a copy of this form for my records, and I have noted the agreed appointment dates.

✓ I understand that I will need to produce some written material as evidence of my progress for the MPhil/PhD Programme by 7 August.

Research Proposal: Structural Dynamics and Asymmetric Pass-Through of Fuel Tax Interventions

Supervisor: Dr. Joao Granj

March 2026

1 Research Summary

This project investigates the impact of temporary fiscal policy on retail fuel markets in the form of excise tax cuts in a special setting characterized by cycling prices. The natural experiment provided by the post-COVID inflation-relief excise tax cut in Australia presents an opportunity to study this phenomenon, which is currently significant in light of the recent global oil price shock.

Oil prices are well-known to follow an asymmetric cycling pattern in Australia, in which industry competitors alternate in undercutting prices until a certain price floor, then a discrete jump-back to a higher price signals the start of another cycle. By employing a Markov Switching Model (Noel, 2007), the study classifies stations into latent pricing regimes of “undercutting,” “stable,” and “relenting” to evaluate whether a major tax cut alters structural competition parameters. Using a Regression Discontinuity Design (RDD) in Time, we measure the extent to which the tax cut, as an exogenous cost-shift, was efficiently passed-through to consumers. Specifically, we ask if the answers to such questions differ across markets endowed with distinct cycling patterns, quantifying the asymmetric effect of regulatory “jawboning” (i.e., regulatory threats to firms in non-legal forms) during the restoration phase.

2 Contextual Background

The 2022 Australian fuel excise intervention serves as a unique natural experiment to study the transmission of fiscal shocks. As documented by the ACCC, Australian petrol pricing is defined by Edgeworth Cycles—a “sawtooth” pattern of sharp hikes and slow decays. Recent international evidence suggests that such “inflation relief” tax cuts often result in incomplete pass-through (Noel, Roach, and Hill, 2024). This research examines whether these cycles, which vary from the deterministic 24-hour cycle in Perth to stochastic coordination in Sydney (Byrne and de Roos, 2019), facilitate or obscure the pass-through of tax changes to consumers.

3 Research Questions

- **RQ1:** Does a large exogenous cost shock induce a structural break in the latent state transition probabilities of Edgeworth cycles (Noel, 2007)?
- **RQ2:** To what extent does the magnitude of pass-through differ between deterministic, stochastic, and non-cycling regimes?

- **RQ3:** What is the relationship between spatial competition density, local market power, and the speed of price adjustment (Deltas, 2008)?
- **RQ4:** Is there a persistent “Rockets and Feathers” effect (Deltas, 2008) during tax restoration, and can it be statistically mitigated by ACCC regulatory signaling (jawboning)?

4 Data Requirements and Sources

Table 1: Data Streams and Primary Sources

Variable	Description	Primary Source
Retail Prices (WA)	Daily prices governed by the “24-hour rule.”	FuelWatch (Data.WA)
Retail Prices (NSW)	Real-time station-level reporting (>2,500 stations).	FuelCheck (Data.NSW)
Wholesale Cost	Daily Terminal Gate Prices (TGP).	AIP / FuelWatch
Tax Policy	Excise rates (44.2 vs 22.1 cpl).	Australian Treasury
Regulatory Signal	ACCC media releases for jawboning intensity.	ACCC Newsroom

References

- Byrne, David P., and Nicolas de Roos. 2019. “Learning to Coordinate: A Study in Retail Gasoline.” *American Economic Review*, 109(2): 591–619.
- Deltas, George. 2008. “Retail Gasoline Price Dynamics and Local Market Power.” *The Journal of Industrial Economics*, 56(3): 613–628.
- Noel, Michael D. 2007. “Edgeworth Price Cycles, Cost-Based Pricing, and Sticky Pricing in Retail Gasoline Markets.” *The Review of Economics and Statistics*, 89(2): 324–334.
- Noel, Michael D., Travis Roach, and Michael Hill. 2024. “Is Inflation Relief Passed-Through? Evidence from Excise Taxes.” *SSRN Working Paper* No. 4944393.